TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Distribute first: multiply the outside number by every term inside.

Get x alone: undo + and − first, then × and ÷. Do the same thing to both sides.

Check: put your answer back in. Both sides equal → correct.

How many solutions? Same x-term and same number: infinite. Same x-term, different number: none. Otherwise: one.

1. What value of p makes the equation $7 - 3(p + 2) = 10 - (2p + 5)$ true?

A. 4

B. −6

C. **−4**

D. No solution

$1 - 3p = 5 - 2p \rightarrow -p = 4 \rightarrow p = -4$

2. Solve the equation shown.

$0.5(4x - 6) = -1.5x + 4$

$x = 2$

$2x - 3 = -1.5x + 4 \rightarrow 3.5x = 7 \rightarrow x = 2$

3. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{2}(8x + 6) = 2(2x + 1) + 2$

No solution

$4x + 3 = 4x + 4$: same x-coefficient, different constants, so $3 = 4$ is never true

4. A triangle has a height of $(x + 5)$ inches and a base of 6 inches. A rectangle has a length of 4 inches and a width of $(x - 1)$ inches. The area of the triangle equals the area of the rectangle. What is the value of x?

$x = 19$

$\frac{1}{2}(6)(x + 5) = 4(x - 1) \rightarrow 3x + 15 = 4x - 4 \rightarrow x = 19$